AJITESH SHUKLA

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EDUCATION

Vellore Institute of Technology, Bhopal, MP — Expected May 2027
B.Tech CSE (Spec. in AI & ML) — CGPA: 9.08/10
Relevant Coursework: Data Structures & Algorithms, OOP, DBMS, Operating Systems, Computer Networks, Machine Learning, Deep Learning
Surya Public School (12th), Lucknow, UP — 85.6% — June 2022
City Montessori School (10th), Lucknow, UP — 91% — June 2020

PROJECTS

India Air Quality Analysis & Prediction — Source Code — May 2026 – Present

- Performed end-to-end data engineering on 29,000+ records across 26 Indian cities, handling missing values via context-aware (city+season) imputation and resolving statistical outliers using percentile-based capping.
- Engineered a custom Composite Pollution Index using weighted normalization of six pollutants, which emerged as the second most influential feature in model predictions per SHAP analysis.
- Built a two-stage ML pipeline: an XGBoost Regressor ($R^2 = 0.93$) to predict continuous AQI values, followed by a Random Forest Classifier (83% accuracy) to categorize air quality into six severity levels.
- Conducted model interpretability analysis using SHAP values to identify feature-level contributions and diagnose model bias toward specific city data anomalies.
- Uncovered actionable insights including a 57% AQI reduction in Delhi during COVID-19 lockdown and identified North Indian geography as a key pollution-trapping factor.

Image Dehazing using Autoencoders — Source Code — Nov 2025 – Present

- Developed a deep learning model using U-Net architecture to effectively remove haziness and restore image clarity.
- Implemented residual learning techniques and spatial attention mechanisms to preserve high-frequency details and optimize reconstruction quality.
- Designed a custom data pipeline with synthetic haze generation to train the network effectively using PyTorch.
- Achieved high scores in image quality metrics (PSNR and SSIM) when evaluated against standard benchmark datasets like RESIDE.

Connect-X AI Agent — Source Code — Jan 2026 – Present

- Engineered an intelligent gaming agent in Python to compete in the Kaggle Connect-X simulation environment.
- Applied game theory concepts, board-state evaluation heuristics, and optimization algorithms to maximize win rates against diverse opponent strategies.
- Implemented the Minimax algorithm with Alpha-Beta pruning to efficiently search future game states.

TECHNICAL SKILLS

- **Languages:** C++, Python, SQL
- **Libraries:** Pandas, NumPy, Scikit-Learn, TensorFlow, Keras, Matplotlib, Seaborn, PyTorch
- **Tools:** Git, GitHub, VS Code, Jupyter Notebook, Google Colab
- **Concepts:** Data Structures (DSA), OOP, DBMS, Machine Learning, Deep Learning, Computer Vision, NLP

CERTIFICATIONS

- **Applied Machine Learning in Python** (University of Michigan) — View Certificate
- **Oracle Cloud Infrastructure 2025 Certified Data Science Professional** — View Certificate
- **Oracle Cloud Infrastructure 2025 Certified AI Foundations Associate** — View Certificate
- **AWS Certified AI Practitioner** — View Certificate

ACHIEVEMENTS

- **Health Hackathon 2026:** Finalist — placed in top 40 out of 600+ teams in hackathon organized by VIT Bhopal and Johns Hopkins University.
- Solved more than 550 problems on LeetCode.
- **Dr. G. Viswanathan Challenge:** Completed 100 Days of Code, strengthening core logic.
- **Teachers Day Challenge:** DSA challenge organized by college to enhance coding skills.